PROPERNESS IN THE GOOD DIAGRAM APPEARS AT A FINITE STAGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5480-5497, following the statement at 5473-5478.

Editorial note. The final step is a construction-reading sentence ('it is immediate from the construction'), anchored to the well-definedness remark; flagged in the notes.

Allowed standard prerequisites (editorial).

- Well-definedness of the good diagram choice. remark-finite-type-gives-well-defined-system (same chapter, cited at ~5482-5486). The choice of i_1 and W is immaterial; we may choose W as a proper fp approximation.
- Proper fp replacement of the limit. lemma-proper-limit-of-proper-finite-presentation (same chapter, cited at ~5488-5489). Closed immersion X -> X' with X' -> Y proper and of finite presentation.
- Relative approximation and eventual properness. lemma-relative-approximation (same chapter, card L26318) and lemma-eventually-proper (same chapter, cited at ~5493-5494). Choose i_3 and a proper morphism W -> Y_{i_3} with X' = Y x_{Y_{i_3}} W.

Source reference. Original equation 0CP8: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5264–5264. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08K1: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2023–2023. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CPA: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5324–5324. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A0W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3137–3137. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09NS: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2859–2859. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 0CPB: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5404–5404. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Lemma 0CPA. If f is proper, then there exists an $i_3 \geq i_0$ such that for $i \geq i_3$ we have f_i is proper.*

Proof. By the discussion in Remark 0CPB the choice of i_1 and W fitting into a diagram as in (0CP8) is immaterial for the truth of the lemma. Thus we choose W as follows. First we choose a closed immersion $X \to X'$ with $X' \to Y$ proper and of finite presentation, see Lemma 0A0W. Then we choose an $i_3 \geq i_2$ and a proper morphism $W \to Y_{i_3}$ such that $X' = Y \times_{Y_{i_3}} W$. This is possible because $Y = \lim_{i \geq i_2} Y_i$ and Lemmas 09NS and 08K1. With this choice of W it is immediate from the construction that for $i \geq i_3$ the algebraic space X_i is a closed subspace of $Y_i \times_{Y_{i_3}} W \subset B_i \times_{B_{i_3}} W$ and hence proper over Y_i. □